UAV Configuration Trade Study

In conducting our trade study to determine the best and most efficient UAV configuration for the VFS DBVF Competition, we decided a stable, multicopter is the best route taking into consideration the mission objective and emphasis on steady hover and transit with a payload weighing 1/5th of our UAV's target weight.

We began by taking inventory of the components we inherited from the prior year, which included three MAD KV200 motors and five of their corresponding 40A ESCs, both of which were rated up to usage with 12S LiPo batteries. We then benchmarked these inherited motors against two competitors of varying price ranges, propeller combinations, and voltage ratings for each multirotor configuration (four-motor, five-motor, and six-motor).

4-Motor Configurations								Total AUW (lb)	Hover Thrust (lbf)	Max Thrust (lbf)	Multiplier
								15	3.75	11.25	3x
Motor	Propeller (in)	Recommended ESC	Throttle at Hover Thrust	Tested Max Thrust (lbf)	Max Current (A)	Max Power (W)	Weight (lb)	Test Voltage at Full Trottle (V)	Compatibility (LiPo)	Cost (USD)	Notes
MAD 5010 EEE V2.0 Drone Motor KV200	17x5.8	AMPX 40A (6S-12S)	52%	11.43	21.62	1029	0.357	47.610	12s	$84	Current Motors 3x
T-Motor MN601-S IP45 Multirotor Fixed Wing UAV Motor KV320	21x6.3in	Flame 70A 6S	50%	12.15	39.4	894	0.543	22.700	6s	$129.99	Requires New ESCs
XRotor M5208 Motor - [UL] 380KV	MSC 18x6.1	XRotor Pro H60A 6S	53%	11.69	44.8	1032	0.345	23	6s	$69.99	Requires New ESCs

5-Motor Configuration								Total AUW (lb)	Hover Thrust (lbf)	Max Thrust (lbf)	Multiplier
								15	3	9	3x
Motor	Propeller (in)	Recommended ESC	Throttle at Hover Thrust	Tested Max Thrust (lbf)	Max Current (A)	Max Power (W)	Weight (lb)	Test Voltage at Full Trottle (V)	Compatibility (LiPo)	Cost (USD)	Notes
MAD 5010 EEE V2.0 Drone Motor KV200	16x5.4	AMPX 40A (6S-12S)	51%	9.7	17.88	855.2	0.357	47.870	12s	$84	Current Motors 3x
XRotor M5208 Motor - [UL] 380KV	17x5.8	XRotor Pro H60A 6S	47%	10	34.7	798.1	0.345	23	6s	$69.99	Requires New ESCs
T-Motor MN501-S IP45 Multirotor Fixed Wing UAV Motor KV360	15x5	Air 40A 6S	45%	10.27	23.11	1200	0.375	48.5	12s	$99.90	

6-Motor Configuration								Total AUW (lb)	Hover Thrust (lbf)	Max Thrust (lbf)	Multiplier
								15	2.5	7.5	3x
Motor	Propeller (in)	Recommended ESC	Throttle at Hover Thrust	Tested Max Thrust (lbf)	Max Current (A)	Max Power (W)	Weight (lb)	Test Voltage at Full Trottle (V)	Compatibility (LiPo)	Cost (USD)	Notes
MAD 5010 EEE V2.0 Drone Motor KV200	24x7.2	AMPX 40A (6S-12S)	58%	8.78	22.31	524.9	0.357	23.680	6s	$84	Current Motors 3x
MAD 5010 EEE V2.0 Drone Motor KV200	16x5.4	AMPX 40A (6S-12S)	47%	9.7	17.88	855.2	0.357	47.870	12s	$84	Current Motors 3x
SunnySky V5210 300KV	20x-	40A	N/A	8.36	26.3	584	0.441	22.2	6s	$86.99	

Figure 1: Motor and Propeller Screening

Collecting this data and interpolating it, we were able to determine criteria for our screening matrix noting the data ranges to develop an objective scoring method.

4-Motor	Motor	Hover Power / Motor (W)	Hover Current / Motor (A)	Total Motor Weight	Additional Motor Cost	Total Tested Max Thrust (lbf)	Max Thrust-to-Weight Ratio	Thrust Margin Above Req	Total Hover Energy for 10 Min (Wh)	Loaded Propulsion Weight Estimate (lb)	Total Additional Cost (Including ESCs)
	MAD 200KV (12S)	199.80	4.18	1.43	$84	45.72	3.05	1.60%	133.2	5.90	$84
	T-Motor 320KV (6S)	179	7.52	2.17	$519.96	48.6	3.24	8.00%	119.33	4.79	$799.92
	Hobbywing 380KV (6S)	185.9	8.06	1.38	$279.96	46.7416	3.12	3.87%	123.93	4.00	$559.92

5-Motor	Motor	Hover Power / Motor (W)	Hover Current / Motor (A)	Total Motor Weight	Additional Motor Cost	Total Tested Max Thrust (lbf)	Max Thrust-to-Weight Ratio	Thrust Margin Above Req	Total Hover Energy for 10 Min (Wh)	Loaded Propulsion Weight Estimate (lb)	Total Additional Cost (Including ESCs)
	MAD 200KV (6S)	158.6	3.3	1.79	$168	48.5	3.23	7.78%	132.17	6.31	$168
	Hobbywing 380KV (6S)	144.9	6.28	1.73	$349.95	50	3.33	11.11%	120.75	4.47	$699.90
	T-Motor 240KV (12S)	210.6	4.33	1.88	$499.50	51.35	3.42	14.11%	175.50	6.40	$499.50

6-Motor	Motor	Hover Power / Motor (W)	Hover Current / Motor (A)	Total Motor Weight	Additional Motor Cost	Total Tested Max Thrust (lbf)	Max Thrust-to-Weight Ratio	Thrust Margin Above Req	Total Hover Energy for 10 Min (Wh)	Loaded Propulsion Weight Estimate (lb)	Total Additional Cost (Including ESCs)
	MAD 200KV (6S)	81.3	3.46	2.14	$252	52.68	3.51	17.07%	81.30	4.58	$301
	MAD 200KV (12S)	125.8	2.62	2.14	$252	58.2	3.88	29.33%	125.80	6.72	$301
	SunnySky 300KV (6S)	98.6	4.45	2.65	$521.94	50.16	3.34	11.47%	98.60	5.08	$570.94

Figure 2: Hover power, current, energy, and purchasing

Putting each of these combinations through our screening matrix with weighted categories for quantitative data recorded from each motor's specification sheet and qualitative characterization for complexity of each approach, we were able to determine that our most optimized configurations are four-motor and six-motor multicopters.

Integration and Schedule Risk								
Score Criteria	Expected Delay	Configuration	Motor Candidate	Remaining Integration Work	Expected Delay (Intregration)	Pessimistic Delay (Integration)	Score	Justification
5	0-1 week	Four Lift Motors	MAD 200KV (12S)	Inspect hardware; verify current weight, payload, CG, battery capacity, and tuning	None	1 week	5	Replacement parts and verification
4	>1-<=2 weeks		T-Motor 320KV (6S)	Configure 6S supply; new mounts; verify 21-inch clearance, distribution, and propulsion compatibility	1-2 weeks	3-4 weeks	4	Mounts, wiring, frame changes, and propulsion testing
			Hobbywing 380KV (6S)	Configure 6S supply; new mounts; verify 18-inch clearance and propulsion compatibility	1-2 weeks	3-4 weeks	4	Mounts, wiring, frame changes, and propulsion testing
3	>2-<=4 weeks	Five Lift Motors	MAD 200KV (12S)	Define geometry; derive custom allocation; schedule simulation, restrained testing, and flight validation	4-6 weeks	6-8 weeks	2	Custom controls or substantial frame/propulsion integration, troubleshooting, reconfiguring comp drone
			Hobbywing 380KV (6S)	Configure 6S supply; new mounts; validate ESC margin and custom allocation	4-6 weeks	6-8 weeks	2	Custom controls or substantial frame/propulsion integration, troubleshooting, reconfiguring comp drone
2	>4-<=6weeks		T-Motor 240KV (12S)	New mounts; derive and validate custom allocation and propulsion compatibility	4-6 weeks	6-8 weeks	2	Custom controls or substantial frame/propulsion integration, troubleshooting, reconfiguring comp drone
		Six Lift Motors	MAD 200KV (6S)	Configure 6S supply; resolve 24-inch prop packaging and structural changes; add sixth branch/output	2-3 weeks	3-4 weeks	3	Mounts, wiring, frame changes, and propulsion testing, reconfiguring comp drone
1	>6 weeks		MAD 200KV (12S)	Six-motor frame; add sixth branch/output; verify distribution and tune standard hex configuration	2-3 weeks	3-4 weeks	3	Replacement parts and verification, reconfiguring comp drone
			SunnySky 300KV (6S)	Configure 6S supply; new mounts; resolve 20-inch packaging; validate propulsion and tune hex	3-4 weeks	4-6 weeks	3	Custom controls or substantial frame/propulsion integration, troubleshooting, reconfiguring comp drone

Figure 3: Quantitative Weighted Schedule Integration and Risk Category

The figure above shows criteria for creating quantitative metrics for the subjective, yet important category of determining how difficult the configuration would be to integrate into our existing schedule. Taking these criteria, as well as further criteria found in figure 4, we then took the top

five performing configurations and moved them to a more in-depth analysis investigating weight and efficiency considerations found in Figure 5.

UAV Motor Configuration Screening Matrix			Four-Motor			Five-Motor			Six-Motor		
	Weight	Preferred Direction	MAD 5010 EEE KV200 (12S)	T-Motor KV320 (6S)	XRotor M5208 Light 380KV (6S)	MAD 5010 EEE KV200 (12S)	XRotor M5208 Light 380KV (6S)	T-Motor KV240 (12S)	MAD 5010 EEE KV200 (6S)	MAD 5010 EEE KV200 (12S)	SunnySky V5210 300KV (6S)
Loaded Propulsion Weight	25%	Lower	5.9	4.79	4	6.31	4.47	6.4	4.58	6.72	5.08
Hover Energy	20%	Lower	133.2	119.3	123.93	132.17	120.75	175.5	81.3	125.8	98.6
Thrust Margin Above Req.	10%	Higher	1.60%	8.00%	3.87%	7.78%	11.11%	14.11%	17.07%	29.33%	11.47%
Control Implementation Readiness	10%	Higher	4	4	4	2	2	2	4	4	4
Additional Purchase Cost	20%	Lower	$84	$799.92	$599.92	$168	$699.90	$499.50	$301	$301	$570.94
Integration & Schedule Risk	15%	-	5	4	4	2	2	2	3	3	3
Technical Ranking		Higher	1.82	3.00	3.73	1.73	3.64	1.55	4.09	2.45	3.09
Overall Score		Higher	3.15	2.40	3.65	2.45	2.90	1.95	3.9	3.00	3.15

Score Criteria	Loaded Propulsion Weight	10-Min Hover Energy	Additional Costs (Motors & ESCs)	Thrust Margin Above 3:1	Score Criteria	Control Implementation Readiness	Configuration	Literature/Studies/Evidence	Score	Justification
5	<=4 lb	<= 90Wh	<=$200	>= 15%	5	Exact proposed configuration implemented and validated against your control requirements	Four Lift Motors	Faessler, Falanga, and Scaramuzza (2017), Thrust Mixing, Saturation, and Body-Rate Control for Accurate Aggressive Quadrotor Flight, provides experimental control validation. ArduPilot quad configuration and Holybro assembly documentation establish implementation precedent; Kakute H743 Wing documentation confirms sufficient outputs and firmware support.	4	Standard Arducopter configuration, documented setup/testing, and sufficient existing FC outputs, still requires validation
4	>4-<=4.5 lb	>90-<=110 Wh	>$200-<=$400	>=10-<15%	4	Standard supported configuration with documented setup and compatible FC output capacity	Five Lift Motors	Lenski (2017), Design, Construction and Operation of a Pentacopter, provides five-motor mixing equations (§3.2.2, Eq. 3.4); Kim et al. (2024), Design and control of a central-rotor type pentacopter, demonstrates a vectored-rotor implementation. Neither validates the proposed five fixed lift-rotor layout. Kakute H743 Wing and ArduPilot custom-matrix documentation support implementation, subject to layout-specific allocation and control validation.	2	Sufficient FC outputs and a custom-mixer implementation path, various configurations have research studies; however, little information on equal distribution pentacopters
3	>4.5-<=5 lb	>110-<=130 Wh	>$400-<=$600	>=5-<10%	3	Custom configuration with completed allocation analysis and representative simulation				
2	>5 -<=6 lb	>130-<=150 Wh	>$600-<=$800	>=2-<5%	2	Implementation path identified; layout-specific allocation/feasibility analysis incomplete	Six Lift Motors	Furrer et al. (2016), RotorS—A Modular Gazebo MAV Simulator Framework, provides hex simulation precedent; Yang et al. (2016), Sliding Mode Control for Hexacopter Stabilization with Motor Failure, demonstrates flight under motor failure using specialized allocation and rotor arrangements. ArduPilot hex configuration and Kakute H743 Wing documentation support standard implementation	4	Standard ArduCopter configuration, documented setup/testing, and sufficient existing FC outputs, still requires validation
1	>6 lb	>150 Wh	>$800	>=0-<2%	1	No established implementation path/demonstrated unresolved control limitation				

Figure 4: Initial Motor Configuration Scoring Matrix

Taking this data and developing weight estimates across the entire propulsion system and comparing each configuration against each other, we were able to determine that the most efficient system is a six-motor aircraft utilizing our existing motor and ESC pairings with 24” propellers run on a single 6S battery. The VFS competition puts emphasis on lightweight designs having a higher points multiplier than their heavier counterparts, leading to this in-depth consideration between configurations.

Taking these findings, we moved to our final dimension of the trade study, which focused on weight, cost, and footprint of various six-motor configurations. We decided to compare a standard Hex X hexacopter, a Y6 tricopter, as well as three designs of alternating motor designs with three equidistant motors above the neutral axis of the arm and three equidistant motors below the neutral axis of the arm with various overlapping propeller percentages (5%, 10%, and 15%).

Structural Weight Calculations for Top Five Candidates

Component / dimension	MAD 200KV quad (12S)	Hobbywing 380KV quad (6S)	MAD 200KV hex (6S)	MAD 200KV hex (12S)	SunnySky V5210 300KV hex (6S)
Propellers (in)	4 × 17 × 5.8	4 × 18 × 6.1	6 × 24 × 7.2	6 × 16 × 5.4	6 × EOLO C20, 20-inch diameter
Center-to-motor distance (in)	12.73	13.44	25	17	21
Total tube length (in)	50.91	53.74	150	102	128.6
Estimated tube weight (lb)	0.63	0.66	1.86	1.26	1.945
Motors, total (lb)	1.437	1.38	2.155	2.155	2.646
Propellers, total (lb)	0.287	0.309	0.794	0.37	0.503
ESCs, total* (lb)	0.203	0.485	0.304	0.304	0.304
Batteries, total (lb)	4.27	2.13	2.13	4.27	2.13
Propulsion-component subtotal (lb)	6.83	4.96	7.24	8.36	7.57
Complete loaded aircraft weight (lb)	12.73	8.96	11.82	15.08	12.65

Note all thrust calculations are for a 15lb AUW

UAV Configuration Screening Matrix			Four-Motor		Six-Motor		
	Weight	Preferred Direction	MAD 200KV quad (12S)	Hobbywing 380KV (6S)	MAD 200KV hex (6S)	MAD 200KV hex (12S)	SunnySky V5210 300KV hex (6S)
Complete Loaded Aircraft Weight	25%	Lower	12.73	8.96	11.82	15.08	12.7
Hover Energy	20%	Lower	133.2	123.93	81.3	125.8	98.6
Thrust Margin Above Req.	10%	Higher	1.60%	3.87%	17.07%	29.33%	11.47%
Control Implementation Readiness	10%	Higher	4	4	4	4	4
Additional Purchase Cost	20%	Lower	$84	$599.92	$301	$301	$570.94
Integration & Schedule Risk	15%	-	5	4	3	3	3
Technical Ranking		Higher	2.73	3.73	4.55	2.45	4.00
Overall Score		Higher	3.65	3.65	4.15	3.00	3.65

Figure 5: Structural weight and second-stage screening

Below, we reweighted the criteria under tighter thresholds, as all remaining configurations are within a threshold for what was previously defined as a four or five on the scoring rubric.

Estimates based on Brazinskas, Prior, and Scanlan (2016), An Empirical Study of Overlapping Rotor Interference for a Small Unmanned Aircraft Propulsion System (3%, 5%, and 8% Thrust Loss Estimated for 5%, 10%, and 15% Prop Interference)

Score Criteria	Loaded Propulsion Weight	10-Min Hover Energy	Additional Costs (Motors & ESCs)	Thrust Margin Above 3:1	Footprint
5	<=10.75 lb	<= 90Wh	<=$200	>= 15%	<4.5 ft
4	>10.75-<=11.25 lb	>90-<=110 Wh	>$200-<=$300	>=10-<15%	>4.5-<=5 ft
3	>11.25-<=11.75 lb	>110-<=130 Wh	>$300-<=$400	>=5-<10%	>5-<=5.5 ft
2	>11.75 -<=12.25 lb	>130-<=150 Wh	>$400-<=$500	>=2-<5%	>5.5-<=6 ft
1	>12.25 lb	>150 Wh	>$500	>=0-<2%	>6 ft

Score Criteria	Control Implementation Readiness	Configuration	Literature/Studies/Evidence	Score	Justification
5	Exact proposed configuration implemented and validated against your control requirements	Hex-X	ArduPilot — Connect ESCs and Motors documents standard hexacopter motor arrangements and motor assignments. The established control architecture and documented motor mixing reduce implementation complexity, making Hex-X a lower-risk baseline for flight-controller integration, tuning, and validation.	4	Standard ArduPilot-supported hexacopter configuration with documented motor allocation and mixing. Requires verification of six ESC outputs, motor order, and flight-controller compatibility.
4	Standard supported configuration with documented setup and compatible FC output capacity	Y6 Coaxial	Experimental Study into Optimal Configuration and Operation of Two-Four Rotor Coaxial Systems for eVTOL Vehicles (2022) experimentally investigates coaxial rotor interference, demonstrating that rotor spacing and operating conditions affect thrust and aerodynamic efficiency. The findings support the need for thrust-stand testing to quantify coaxial performance losses and verify the Y6 configuration's ability to meet the required thrust-to-weight ratio.	4	Standard ArduPilot-supported Y6 configuration with established six-motor mixing. Coaxial aerodynamic interference requires tuning and validation but does not inherently require custom motor allocation.
3	Custom configuration with completed allocation analysis and representative simulation	Alt. Hex 5%	Zhu et al. (2023) and Brazinskas et al. (2016) experimentally demonstrate that rotor overlap and vertical separation influence thrust performance and aerodynamic efficiency. These studies support investigating staggered rotor configurations but highlight the need for configuration-specific testing to quantify interference losses, validate thrust margins, and assess control effectiveness.	3	Conventional hex motor allocation may be usable, but vertically staggered overlapping rotors introduce aerodynamic coupling that must be evaluated through simulation and testing. Potential custom motor mixing needed
2	Implementation path identified; layout-specific allocation/feasibility analysis incomplete	Alt. Hex 10%	Zhu et al. (2023) and Brazinskas et al. (2016) experimentally demonstrate that rotor overlap and vertical separation influence thrust performance and aerodynamic efficiency. These studies support investigating staggered rotor configurations but highlight the need for configuration-specific testing to quantify interference losses, validate thrust margins, and assess control effectiveness.	3	Conventional hex control architecture may remain applicable, but increased overlap creates greater uncertainty in thrust response and control effectiveness. Requires representative validation. Potential custom motor mixing needed
1	No established implementation path/demonstrated unresolved control limitation	Alt. Hex 15%	Zhu et al. (2023) and Brazinskas et al. (2016) experimentally demonstrate that rotor overlap and vertical separation influence thrust performance and aerodynamic efficiency. These studies support investigating staggered rotor configurations but highlight the need for configuration-specific testing to quantify interference losses, validate thrust margins, and assess control effectiveness.	3	Potentially greater rotor interaction and control-authority uncertainty. Requires thrust characterization, control-allocation verification, and representative simulation before flight. Potential custom motor mixing needed

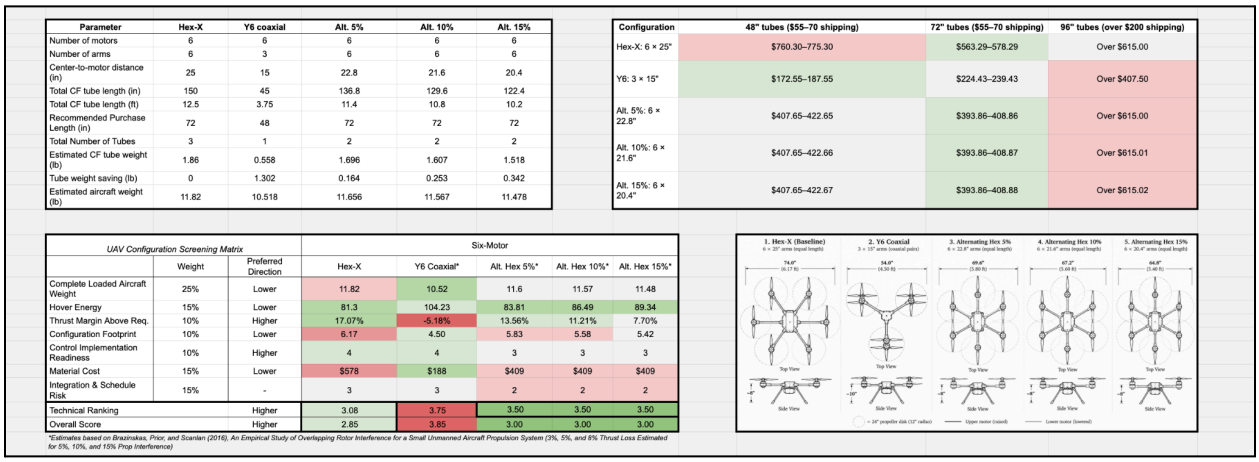
Integration and Schedule Risk

Score Criteria	Expected Delay	Configuration	Remaining Integration Work	Expected Delay (Integration)	Pessimistic Delay (Integration)	Score	Justification
5	0-1 week	Hex-X	Fabricate six-arm frame; integrate 6S supply, six ESCs, and flight-controller outputs; verify 24-inch prop clearance; conduct motor and hover testing.	2-3 weeks	3-4 weeks	3	Standard hex configuration minimizes control development, but six-motor wiring, structural fabrication, and propulsion validation require approximately 2–3 weeks.
4	>1-<=2 weeks	Y6 Coaxial	Fabricate three-arm coaxial frame; design upper/lower motor mounts; establish rotor spacing; validate thrust loss, vibration, and thermal performance.	2-3 weeks	4-6 weeks	3	Standard Y6 control architecture reduces software risk, but coaxial propulsion testing and mounting integration add uncertainty.
3	>2-<=4 weeks	Alt. Hex 5%	Design staggered motor mounts; verify disk overlap and vertical clearance; conduct interference tests; validate control response and hover stability.	4-6 weeks	6-8 weeks	2	Nonstandard geometry requires additional fabrication, propulsion characterization, and flight validation. Estimated 4–6 weeks.
2	>4-<=6weeks	Alt. Hex 10%	Same as 5%, with greater emphasis on aerodynamic interaction, structural clearance, and thrust-margin verification.	4-6 weeks	6-8 weeks	2	Increased rotor interaction uncertainty raises the likelihood of redesign and repeated testing. Estimated 4–6 weeks.
1	>6 weeks	Alt. Hex 15%	Same as 10%, plus investigation of possible thrust shortfall relative to the 3:1 requirement.	4-6 weeks	6-8 weeks	2	Highest propulsion feasibility risk among the alternating configurations under the current assumptions; may require geometric redesign.

Figure 6: Six-Motor Configuration Scoring Matrix Criteria

Scoring each configuration under these criteria, as well as calculating in depth the effective structural cost of materials for each scenario, we were able to determine that while the Y6 Coaxial design is lightest, cheapest, and smallest, it fails to achieve the basic technical specifications of a 3:1 thrust-to-weight ratio, so it was eliminated as a potential option. For the remaining candidates, each of the Alternating Hex designs scored equal technical scores with the Hex-X scoring noticeably lower; however, considering the configurations holistically with integration, schedule risk, and existing literature, the Alternating Hex designs scored only slightly higher than the Hex-X.

Because both designs (Alt. Hex and Hex-X) are relatively similar and resemble a standard hexacopter with six equal-length arms, we have decided to move forward with both designs, with the Hex-X as essentially the target and the Alternating Hex 5-15% as the reach goal, considering the criteria differentiating the two configurations is our lack of knowledge surrounding specific concept integration and validation. We will begin concept validation testing to prove the Alternating Hex concept with ArduPilot control software at 5%, 10%, and 15% overlap and reserve the Hex-X symmetric hexacopter as a failsafe option.



Parameter	Hex-X	Y6 coaxial	Alt. 5%	Alt. 10%	Alt. 15%
Number of motors	6	6	6	6	6
Number of arms	6	3	6	6	6
Center-to-motor distance (in)	25	15	22.8	21.6	20.4
Total CF tube length (in)	150	45	136.8	129.6	122.4
Total CF tube length (ft)	12.5	3.75	11.4	10.8	10.2
Recommended Purchase Length (in)	72	48	72	72	72
Total Number of Tubes	3	1	2	2	2
Estimated CF tube weight (lb)	1.86	0.558	1.696	1.607	1.518
Tube weight saving (lb)	0	1.302	0.164	0.253	0.342
Estimated aircraft weight (lb)	11.82	10.518	11.656	11.567	11.478

Configuration	48" tubes ($55–70 shipping)	72" tubes ($55–70 shipping)	96" tubes (over $200 shipping)
Hex-X: 6 × 25"	$760.30–775.30	$563.29–578.29	Over $615.00
Y6: 3 × 15"	$172.55–187.55	$224.43–239.43	Over $407.50
Alt. 5%: 6 × 22.8"	$407.65–422.65	$393.86–408.86	Over $615.00
Alt. 10%: 6 × 21.6"	$407.65–422.66	$393.86–408.87	Over $615.01
Alt. 15%: 6 × 20.4"	$407.65–422.67	$393.86–408.88	Over $615.02

UAV Configuration Screening Matrix			Six-Motor				
	Weight	Preferred Direction	Hex-X	Y6 Coaxial*	Alt. Hex 5%*	Alt. Hex 10%*	Alt. Hex 15%*
Complete Loaded Aircraft Weight	25%	Lower	11.82	10.52	11.6	11.57	11.48
Hover Energy	15%	Lower	81.3	104.23	83.81	86.49	89.34
Thrust Margin Above Req.	10%	Higher	17.07%	-5.18%	13.56%	11.21%	7.70%
Configuration Footprint	10%	Lower	6.17	4.50	5.83	5.58	5.42
Control Implementation Readiness	10%	Higher	4	4	3	3	3
Material Cost	15%	Lower	$578	$188	$409	$409	$409
Integration & Schedule Risk	15%	-	3	3	2	2	2
Technical Ranking		Higher	3.08	3.75	3.50	3.50	3.50
Overall Score		Higher	2.85	3.85	3.00	3.00	3.00

"Estimates based on Brazinskas, Prior, and Scanlan (2016), An Empirical Study of Overlapping Rotor Interference for a Small Unmanned Aircraft Propulsion System (3%, 5%, and 6% Thrust Loss Estimated for 5%, 10%, and 15% Prop Interference)

Figure 7: Six-Motor Configuration Scoring Matrix Results